Osher Azulay

azulayosher@gmail.com LinkedIn GitHub Website Scholar

Education

Ph.D., Mechanical Engineering, Tel Aviv University 2020–2024
Advisor: Dr. Avishai Sintov
Research Area: Learning in-hand perception and manipulation with adaptive robotic hands

M.Sc., Mechanical Engineering, Ben Gurion University 2018–2020
Advisor: Prof. Amir Shapiro
Fast-Track Program
Thesis: Wheel loader scooping controller using deep reinforcement learning

B.Sc., Mechanical Engineering, Ben Gurion University 2015–2019
Advisor: Prof. Amir Shapiro

Employment

Fulbright Postdoctoral Research Fellow 2025–present
CSE, University of Michigan, Ann Arbor
Advisor: Prof. Stella X. Yu

Visiting Scholar Winter 2025
AUTOLab, EECS, University of California, Berkeley

Visiting Student Summer 2023
Robot Learning Lab, CS, Rutgers University
– Studied visual-tactile-based object insertion and sim-to-real transfer.

Robotics Engineer Summer 2022
Unlimited Robotics
– Explored ROS2 control for a dual-arm robot, from simulation to hardware integration.

Research Assistant 2016–2018
BGU Robotics Control Lab, MechEng, Ben-Gurion University
– Provided technical support for ROS platforms, including arms and mobile robots.

Teaching Experience

Advanced Topics in Computer Vision (EECS 542) Winter 2026
LEO Adjunct Lecturer, *EECS, University of Michigan*

Robotics and Control Lab Spring 2021–24
Designed and created course material, *Mech Eng., Tel-Aviv University*

Introduction to Control Theory Fall 2020–24
Teaching Assistant, *Mech Eng., Tel-Aviv University*

Introduction to Electrical Engineering Spring 2019
Teaching Assistant, *Mech Eng., Ben-Gurion University*

C Programming Fall 2019
Teaching Assistant, *Mech Eng., Ben-Gurion University*

Introduction to Mechanical Engineering Fall 2018
Lab Instructor, *Mech Eng., Ben-Gurion University*

Honors & Recognition

2026 | Honorable Mention for a talk on humanoid fall safety, ScienceAbroad Applied Science Symposium.

2024 | Awarded the Fulbright Postdoctoral Fellowship.
Awarded the KLA Scholarships for PhD excellence.
Awarded the prestigious ME Graduate Research Award.
Received travel award from the Center for AI and Data Science at Tel Aviv University.

2023 | Awarded the KLA Scholarships for PhD excellence.
Received Honorable Mention from the Dean for Excellence in Teaching.
Received travel award from the IEEE Robotics and Automation Society.

2022 | Awarded the Prof. N. Levtzion Scholarships for outstanding doctoral students.

2019 | Awarded Certificate of Achievement for outstanding academic performance.

2018 | Awarded Certificate of Achievement for outstanding academic performance.

Invited Talks & Presentations

2026 | "Unified Fall Safety for Humanoid Robots," ScienceAbroad Global Israeli Applied Science Symposium.

2025 | Invited Talk, ROAM Lab, Department of Mechanical Engineering, Columbia University.

2024 | Invited talk at the Annual Meeting for Motion Control and Automation.
Invited Colloquium Talk, Department of Computer Science, Bar-Ilan University.
Invited Seminar Talk, Department of Mechanical Engineering, Technion – Israel Institute of Technology.

2022 | Invited talk at the Annual Meeting for Motion Control and Automation.

Manuscripts Under Review

[*] Equal contribution.

2026

1. **Azulay, O.**[*], Hsieh, C.-L.[*] & Yu, S. X. Test-Time Motion Steering for Perceptive Humanoid Fall Recovery. *arXiv:2610.30061, under review* (2026). Under review
2. Curtis, N.[*], **Azulay, O.**[*] & Sintov, A. Embodiment-Agnostic Navigation Policy Trained with Visual Demonstrations. *Under review* (2026). Under review
3. Scheffer, D., **Azulay, O.** & Yu, S. X. EgoIntercept: Egocentric Object Interception with a Quadruped Robot. *Under review* (2026). Under review
4. Liu, S., Zhang, Z., Li, Z., Lin, Y., Lin, R., Liu, X., Nguyen, K., Gao, Z., Liao, G., Mo, J., Jiang, L., Yang, J., **Azulay, O.**, Yu, S. X., Yu, P. S. & Yan, Q. Physical AI: A Comprehensive Survey of Enabling Technologies From A Systems Perspective. *Under review at the Journal of Artificial Intelligence Research (JAIR)* (2026). Under review

Publications

2026

5. **Azulay, O.**, Xu, Z., Scheffer, A. & Yu, S. X. Unified Perceptive Fall Safety for Humanoids. *IEEE Robotics and Automation Letters, accepted* (2026). Journal

2025

6. **Azulay, O.**[*], Kondap, K.[*], Drake, J.[*], Xie, S., Li, H., Chitta, S. & Goldberg, K. MOTORCYCLE 1.0: Automating Bimanual Cable Routing Around Fixtures on the NIST Task Board. *IEEE International Conference on Automation Science and Engineering (CASE)* (2025). Conference
7. **Azulay, O.**, Ramesh, D. M., Curtis, N. & Sintov, A. Visuotactile-Based Learning for Insertion With Compliant Hands. *IEEE Robotics and Automation Letters* (2025). *(Also presented at IROS 2025.)* Journal + conf.
8. Yu, J.[*], Shivakumar, N.[*], Sumedh, V.[*], Zhang, J., Ransing, E., **Azulay, O.** & Goldberg, K. HANDLOOM 3.0: Interactive Bi-Directional Cable Tracing Amid Clutter. *ICRA Workshop on Reflections on Representations and Manipulating Deformable Objects (RMDO)* (2025). Workshop

2024

9. Arolovitch, J.[*], **Azulay, O.**[*] & Sintov, A. Kinesthetic-based In-Hand Object Recognition with an Underactuated Robotic Hand. *IEEE International Conference on Robotics and Automation (ICRA)* (2024). Conference
10. **Azulay, O.**, Curtis, N., Sokolovsky, R., Levitski, G., Slomovik, D., Lilling, G. & Sintov, A. AllSight: A Low-Cost and High-Resolution Round Tactile Sensor with Zero-Shot Learning Capability. *IEEE Robotics and Automation Letters* (2024). *(Also presented at ICRA 2024.)* Journal + conf.
11. **Azulay, O.**[*], Mizrahi, A.[*], Curtis, N.[*] & Sintov, A. Augmenting Tactile Simulators with Real-like and Zero-Shot Capabilities. *IEEE International Conference on Robotics and Automation (ICRA)* (2024). Conference
12. **Azulay, O.** & Sintov, A. AllSight: Advancing Optical Tactile Sensing and Sim-to-Real Learning for Dexterous Robotic Manipulation. *ICRA Workshop on Robot Embodiment through Visuo-Tactile Perception (ViTac)* (2024). Workshop
13. Weinberg, A. I., Shirizly, A., **Azulay, O.** & Sintov, A. Survey of Learning-Based Approaches for Robotic In-Hand Manipulation. *Frontiers in Robotics and AI* (2024). Journal

2023

14. **Azulay, O.**, Ben-David, I. & Sintov, A. Learning Haptic-Based Object Pose Estimation for In-Hand Manipulation Control With Underactuated Robotic Hands. *IEEE Transactions on Haptics* (2023). Journal
15. **Azulay, O.**, Monastirsky, M. & Sintov, A. Haptic-based and SE(3)-aware object insertion using compliant hands. *IEEE Robotics and Automation Letters* (2023). *(Also presented at ICRA 2023 and the Israeli Conference on Robotics (ICR) 2023.)* Journal + conf.
16. Monastirsky, M., **Azulay, O.** & Sintov, A. Learning to Throw With a Handful of Samples Using Decision Transformers. *IEEE Robotics and Automation Letters* (2023). *(Also presented at IROS 2023.)* Journal + conf.

2021

17. **Azulay, O.** & Shapiro, A. Wheel Loader Scooping Controller Using Deep Reinforcement Learning. *IEEE Access* (2021). Journal
18. Bamani, E., **Azulay, O.**, Gurevich, A. & Sintov, A. Open-Sourcing Generative Models for Data-driven Robot Simulations. *Data-Centric AI Workshop, NeurIPS* (2021). Workshop

Professional Service

Conference reviewer: ICRA (2024, 2026–2027), IROS (2020, 2025–2026).

Journal reviewer: IEEE RA-L (2023–2024, 2026), IEEE TPAMI (2026), ACM THRI (2026).